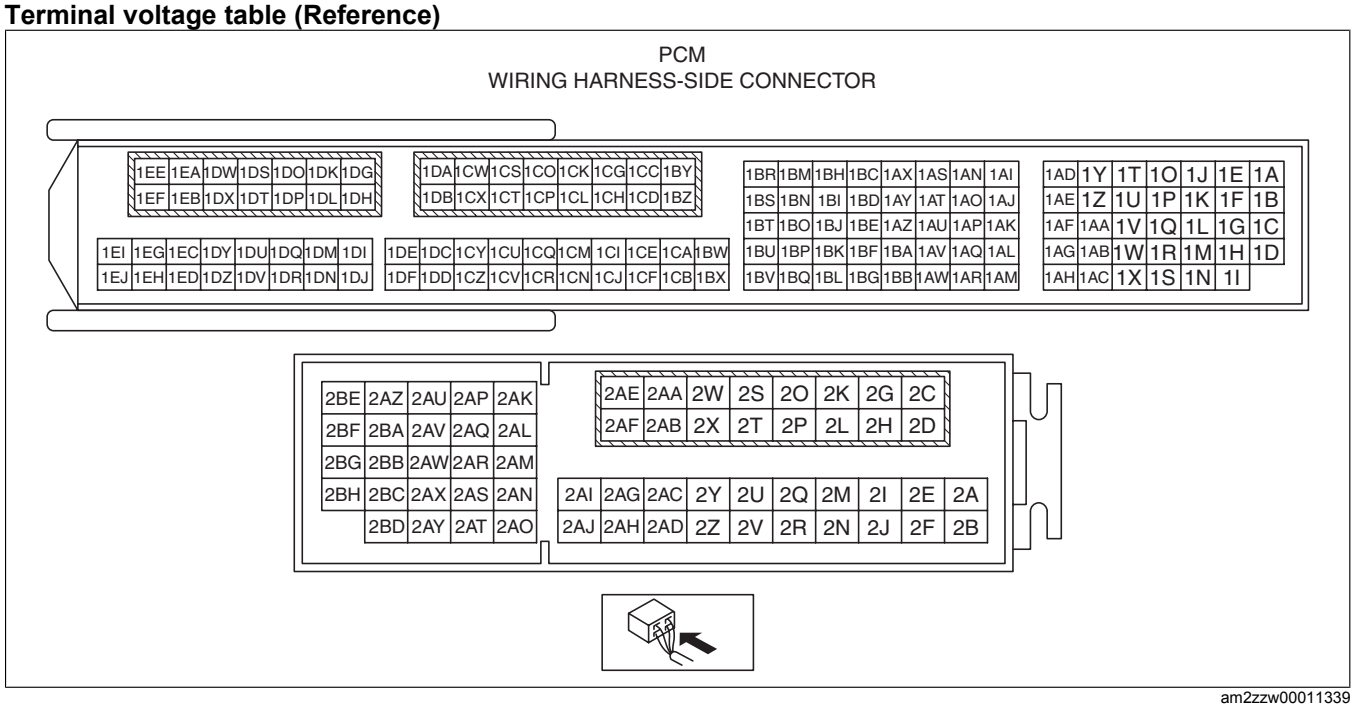
PCM INSPECTION [SKYACTIV-D 1.5]

id0140q2113900

Note

- Because the PCM uses a waterproof connector, the inspection for the voltage/wave pattern cannot be performed. The following values are for reference.

Terminal voltage table (Reference)



am2zzw00011339

—: Not applicable

Terminal	Signal	Connected to	Test condition	Voltage (V)	Inspection item
1A	CMP (G+)	CMP sensor	(See CMP (G+) signal.)		• CMP sensor • Related wiring harness
1B	CMP (G−)	CMP sensor	Switch ignition off	0	• CMP sensor • Related wiring harness
			Switch ignition ON (engine off)	Approx. 0.02	
1C*1	CAN2_L	CAN system related modules	Because this terminal is for CAN, good/no good judgment by terminal voltage is not possible.		• Related wiring harness
1D	GND	Exhaust gas pressure sensor No.2	Switch ignition ON (engine off)	Approx. 0.02	• Related wiring harness
1E	CKP (NE+)	CKP sensor	(See CKP (NE+) signal.)		• CKP sensor • Related wiring harness
1F	CKP (NE−)	CKP sensor	Switch ignition ON (engine off)	Approx. 0.01	• CKP sensor • Related wiring harness
1G*1	CAN2_H	CAN system related modules	Because this terminal is for CAN, good/no good judgment by terminal voltage is not possible.		• Related wiring harness
1H	A/F (+)	A/F sensor	Idle (after warm up)	Approx. 3.04	• A/F sensor • Related wiring harness
1I	A/F (−)	A/F sensor	Idle (after warm up)	Approx. 2.50	• A/F sensor • Related wiring harness
1J	Constant voltage (Vref)	CKP sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1K	Constant voltage (Vref)	CMP sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1L	GND	Sensor shield	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1M	—	—	—	—	—

Terminal	Signal	Connected to	Test condition	Voltage (V)	Inspection item
1N	Boost air temperature	Boost air temperature sensor	Switch ignition ON (engine off)	Approx. 3.14	• Boost air temperature sensor • Related wiring harness
1O*3	Generator output voltage	Generator	(See Generator output voltage signal.)		• Generator • Related wiring harness
1P	Blow-by heater relay	Blow-by heater relay	Switch ignition ON (engine off)	Approx. 0.09	• Blow-by heater relay • Related wiring harness
			Idle (after warm up)	Approx. 0.10	
1Q	LIN	Glow control module, engine oil level sensor*5	Because this terminal is for CAN, good/no good judgment by terminal voltage is not possible.		• Glow control module • engine oil level sensor*5 • Related wiring harness
1R	—	—	—	—	—
1S	—	—	—	—	—
1T	—	—	—	—	—
1U	—	—	—	—	—
1V	—	—	—	—	—
1W	Engine oil temperature	Engine oil temperature sensor	Switch ignition ON (engine off)	Approx. 2.04	• Engine oil temperature sensor • Related wiring harness
1X	Exhaust gas pressure	Exhaust gas pressure sensor No.1	Switch ignition ON (engine off)	Approx. 0.96	• Exhaust gas pressure sensor No.1 • Related wiring harness
			Idle (after warm up)	1.06—1.08	
1Y	—	—	—	—	—
1Z	—	—	—	—	—
1AA	Constant voltage (Vref)	Exhaust gas pressure sensor No.2	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1AB	Engine oil pressure	Engine oil pressure sensor	Switch ignition ON (engine off)	Approx. 0.512	• Engine oil pressure sensor • Related wiring harness
			Idle (after warm up)	1.14—1.15	
1AC	Exhaust gas pressure	Exhaust gas pressure sensor No.2	Switch ignition ON (engine off)	Approx. 0.50	• Exhaust gas pressure sensor No.2 • Related wiring harness
			Idle (after warm up)	0.59—0.60	
1AD	A/F sensor heater control	A/F sensor heater	(See A/F sensor heater control signal.)		• A/F sensor heater • Related wiring harness
1AE	—	—	—	—	—
1AF	Constant voltage (Vref)	Fuel pressure sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1AG	Fuel pressure	Fuel pressure sensor	Switch ignition ON (engine off)	Approx. 0.52	• Fuel pressure sensor • Related wiring harness
			Idle (after warm up)	1.04—1.06	
1AH	GND	Fuel pressure sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1AI	LP-EGR control valve (LPEGR−)	LP-EGR control valve	Switch ignition ON (engine off)	Approx. 12.67	• LP-EGR control valve • Related wiring harness
			(See LP-EGR control valve (LPEGR-).)		
1AJ	Exhaust shutter valve (ESV−)	Exhaust shutter valve	Switch ignition ON (engine off)	Approx. 12.66	• Exhaust shutter valve • Related wiring harness
			(See Exhaust shutter valve (ESV-).)		
1AK	Constant voltage (Vref)	Exhaust shutter valve position sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1AL	Exhaust shutter valve position	Exhaust shutter valve position sensor	Switch ignition ON (engine off)	Approx. 4.28	• Exhaust shutter valve position sensor • Related wiring harness
			Idle (after warm up)	Approx. 4.06	

Terminal	Signal	Connected to	Test condition	Voltage (V)	Inspection item
1AM	GND	Exhaust shutter valve position sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1AN	LP-EGR control valve (LPEGR+)	LP-EGR control valve	Switch ignition ON (engine off)	Approx. 12.62	• LP-EGR control valve • Related wiring harness
			Idle	Approx. 14.29	
1AO	Exhaust shutter valve (ESV+)	Exhaust shutter valve	Switch ignition ON (engine off)	Approx. 12.61	• Exhaust shutter valve • Related wiring harness
			Idle	Approx. 14.28	
1AP	Constant voltage (Vref)	LP-EGR control valve position sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1AQ	LP-EGR control valve position	LP-EGR control valve position sensor	Switch ignition ON (engine off)	Approx. 1.20	• LP-EGR control valve position sensor • Related wiring harness
			Idle (after warm up)	1.29—1.31	
1AR	GND	LP-EGR control valve position sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1AS	Intake shutter valve (ISV-)	Intake shutter valve	Switch ignition ON (engine off)	Approx. 12.54	• Intake shutter valve • Related wiring harness
			(See Intake shutter valve (ISV-) signal.)		
1AT	—	—	—	—	—
1AU	Constant voltage (Vref)	Intake shutter valve position sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1AV	Intake shutter valve position	Intake shutter valve position sensor	Switch ignition ON (engine off)	Approx. 4.17	• Intake shutter valve position sensor • Related wiring harness
			Idle (after warm up)	0.71—0.72	
1AW	GND	Intake shutter valve position sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1AX	Intake shutter valve (ISV+)	Intake shutter valve	Switch ignition ON (engine off)	Approx. 12.51	• Intake shutter valve • Related wiring harness
			Idle	Approx. 14.15	
1AY*4	Generator field coil control	Generator	(See Generator field coil control signal.)		• Generator • Related wiring harness
1AZ	Constant voltage (Vref)	MAP sensor, boost air temperature sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1BA	MAP	MAP sensor	Switch ignition ON (engine off)	Approx. 1.63	• MAP sensor • Related wiring harness
			Idle (after warm up)	1.50—1.52	
1BB	GND	MAP sensor, boost air temperature sensor	Switch ignition ON (engine off)	Approx. 0.02	• Related wiring harness
1BC	HP-EGR control valve (EGRH−)	HP-EGR control valve	Switch ignition ON (engine off)	Approx. 12.89	• HP-EGR control valve • Related wiring harness
			Idle (when cold)	Approx. 13.00	
			Idle (after warm up)	Approx. 13.25	
1BD	—	—	—	—	—

Terminal	Signal	Connected to	Test condition	Voltage (V)	Inspection item
1BE	Constant voltage (Vref)	HP-EGR control valve position sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1BF	HP-EGR control valve position	HP-EGR control valve position sensor	Switch ignition ON (engine off)	Approx. 1.21	• HP-EGR control valve • Related wiring harness
			Idle (after warm up)	Approx. 2.06	
1BG	GND	HP-EGR control valve position sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1BH	HP-EGR control valve (EGRH+)	HP-EGR control valve	Switch ignition ON (engine off)	Approx. 12.78	• HP-EGR control valve • Related wiring harness
1BI	Constant voltage (Vref)	Speed sensor assy (turbocharger with variable turbine geometry)	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
			Idle (after warm up)	Approx. 5.10	
1BJ	Constant voltage (Vref)	Actuator position sensor (turbocharger with variable turbine geometry)	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1BK	Actuator position	Actuator position sensor (turbocharger with variable turbine geometry)	Switch ignition ON (engine off)	Approx. 0.58	• Actuator position sensor (turbocharger with variable turbine geometry) • Related wiring harness
			Idle (after warm up)	Approx. 3.38	
1BL	GND	Actuator position sensor (turbocharger with variable turbine geometry)	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1BM	Coolant control valve (3WAY−)	Coolant control valve	Switch ignition ON (engine off)	Approx. 12.72	• Coolant control valve • Related wiring harness
			Idle (when cold)	Approx. 14.34	
			Idle (after warm up)	Approx. 14.33	
1BN	—	—	—	—	—
1BO	Constant voltage (Vref)	Coolant control valve position sensor	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1BP	Coolant control valve position	Coolant control valve position sensor	Switch ignition ON (engine off)	Approx. 4.53	• Coolant control valve • Related wiring harness
			Idle (after warm up)	1.60—1.62	
1BQ	GND	Coolant control valve position sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1BR	Coolant control valve (3WAY+)	Coolant control valve	Switch ignition ON (engine off)	Approx. 12.68	• Coolant control valve • Related wiring harness
1BS	—	—	—	—	—
1BT	Constant voltage (Vref)	Engine oil temperature sensor, engine oil pressure sensor	Switch ignition ON (engine off)	Approx. 5.02	• Related wiring harness
1BU*4	Generator field coil control	Generator	(See Generator output voltage signal.)		• Generator • Related wiring harness
1BV	GND	Engine oil temperature sensor, engine oil pressure sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness

Terminal	Signal	Connected to	Test condition	Voltage (V)	Inspection item
1BW	Exhaust gas temperature	Exhaust gas temperature sensor No.3	Switch ignition ON (engine off)	Approx. 4.78	• Exhaust gas temperature sensor No. 3 • Related wiring harness
1BX	GND	Exhaust gas temperature sensor No.3	Under any condition	Approx. 0.01	• Related wiring harness
1BY	Suction control valve	Suction control valve	Switch ignition ON (engine off)	Approx. 0.01	• Suction control valve • Related wiring harness
			Idle (after warm up)	0.44—0.49	
1BZ	GND	GND	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CA	Exhaust gas temperature	Exhaust gas temperature sensor No.2	Switch ignition ON (engine off)	Approx. 4.70	• Exhaust gas temperature sensor No. 2 • Related wiring harness
1CB	GND	Exhaust gas temperature sensor No.2	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CC	Suction control valve	Suction control valve	Switch ignition ON (engine off)	Approx. 0.01	• Suction control valve • Related wiring harness
			(See Suction control valve signal.)		
1CD	GND	GND	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CE	Exhaust gas temperature	Exhaust gas temperature sensor No.4	Switch ignition ON (engine off)	Approx. 4.96	• Exhaust gas temperature sensor No. 4 • Related wiring harness
1CF	GND	Exhaust gas pressure sensor No.1	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CG*3	Generator field coil control	Generator	(See Generator field coil control signal.)		• Generator • Related wiring harness
1CH	Turbocharger solenoid valve	Turbocharger solenoid valve	Switch ignition ON (engine off)	Approx. 12.38	• Turbocharger solenoid valve • Related wiring harness
			(See Turbocharger solenoid valve.)		
1CI	Exhaust gas temperature	Exhaust gas temperature sensor No.1	Switch ignition ON (engine off)	Approx. 4.98	• Exhaust gas temperature sensor No. 1 • Related wiring harness
1CJ	Constant voltage (Vref)	Exhaust gas pressure sensor No.1	Switch ignition ON (engine off)	Approx. 5.01	• Related wiring harness
1CK	—	—	—	—	—
1CL	—	—	—	—	—
1CM	—	—	—	—	—
1CN	GND	Exhaust gas temperature sensor No.1	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CO	Engine oil control	Engine oil solenoid valve	Switch ignition ON (engine off)	Approx. 12.37	• Engine oil solenoid valve • Related wiring harness
			(See Engine oil control signal.)		
1CP	Electric water pump	Electric water pump	Switch ignition ON (engine off)	Approx. 0.02	• Electric water pump • Related wiring harness
1CQ	IAT (No.2)	IAT sensor No.2	Switch ignition ON (engine off)	Approx. 2.52	• IAT sensor No.2 • Related wiring harness
1CR	GND	IAT sensor No.2	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CS	Fuel injection control (+)	Fuel injector No.3	(See Fuel injection control (+) signal.)		• Fuel injector No.3 • Related wiring harness

Terminal	Signal	Connected to	Test condition	Voltage (V)	Inspection item
1CT	Fuel injection control (+)	Fuel injector No.2	(See Fuel injection control (+) signal.)		• Fuel injector No.2 • Related wiring harness
1CU	Fuel temperature	Fuel temperature sensor	Switch ignition ON (engine off)	Approx. 1.97	• Fuel temperature sensor • Related wiring harness
1CV	GND	Fuel temperature sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1CW	Fuel injection control (−)	Fuel injector No.2	(See Fuel injection control (-) signal.)		• Fuel injector No.2 • Related wiring harness
1CX	—	—	—	—	—
1CY	ECT	ECT sensor	Switch ignition ON (engine off)	Approx. 2.02	• ECT sensor • Related wiring harness
1CZ	GND	ECT sensor	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1DA	Fuel injection control (−)	Fuel injector No.3	(See Fuel injection control (-) signal.)		• Fuel injector No.3 • Related wiring harness
1DB	—	—	—	—	—
1DC	—	—	—	—	—
1DD	—	—	—	—	—
1DE	—	—	—	—	—
1DF	GND	Sensor shield	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1DG	PRD−	Fuel pressure relief valve	Switch ignition ON (engine off)	Approx. 2.58	• Fuel pressure relief valve • Related wiring harness
			Idle (after warm up)	Approx. 2.65	
1DH	Battery voltage	Main relay	Switch ignition ON (engine off)	B+	• Related wiring harness
1DI	—	—	—	—	—
1DJ	—	—	—	—	—
1DK	PRD+	Fuel pressure relief valve	Switch ignition ON (engine off)	Approx. 2.58	• Fuel pressure relief valve • Related wiring harness
			Idle (after warm up)	Approx. 2.66	
1DL	Battery voltage	Main relay	Switch ignition ON (engine off)	B+	• Related wiring harness
1DM	—	—	—	—	—
1DN	—	—	—	—	—
1DO	—	—	—	—	—
1DP	—	—	—	—	—
1DQ	GND	Sensor shield	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1DR	—	—	—	—	—
1DS	—	—	—	—	—
1DT	—	—	—	—	—
1DU	—	—	—	—	—
1DV	GND	Exhaust gas temperature sensor No.4	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1DW	Fuel injection control (−)	Fuel injector No.4	(See Fuel injection control (-) signal.)		• Fuel injector No.4 • Related wiring harness
1DX	—	—	—	—	—
1DY	GND	Sensor shield	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1DZ	GND	Speed sensor assy (turbocharger with variable turbine geometry)	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1EA	Fuel injection control (−)	Fuel injector No.1	(See Fuel injection control (-) signal.)		• Fuel injector No.1 • Related wiring harness
1EB	—	—	—	—	—
1EC	GND	Sensor shield	Switch ignition ON (engine off)	Approx. 0.01	• Related wiring harness
1ED	—	—	—	—	—

Terminal	Signal	Connected to	Test condition		Voltage (V)	Inspection item
1EE	Fuel injection control (+)	Fuel injector No.1	(See Fuel injection control (+) signal.)			• Fuel injector No.1 • Related wiring harness
1EF	Fuel injection control (+)	Fuel injector No.4	(See Fuel injection control (+) signal.)			• Fuel injector No.4 • Related wiring harness
1EG	—	—	—		—	—
1EH	—	—	—		—	—
1EI	—	—	—		—	—
1EJ	TURBINE ROTATION SPEED SIGNAL	Speed sensor assy (turbocharger with variable turbine geometry)	(See TURBINE ROTATION SPEED SIGNAL.)			• Speed sensor assy (turbocharger with variable turbine geometry) • Related wiring harness
2A	—	—	—		—	—
2B*2	Neutral position	Neutral switch No. 1	Shift lever is at neutral position		Below 1.0	• Neutral switch No.1 • Related wiring harness
			Shift lever is not at neutral position		B+	
2C	HS CAN_H	CAN system related modules	Because this terminal is for CAN, good/no good judgment by terminal voltage is not possible.			• Related wiring harness
2D	HS CAN_L	CAN system related modules	Because this terminal is for CAN, good/no good judgment by terminal voltage is not possible.			• Related wiring harness
2E	Check connector	Check connector	Switch ignition ON (engine off)		Approx. 12.83	• Check connector • Related wiring harness
2F	—	—	—		—	—
2G*2	Neutral switch No.2	Neutral switch No. 2	Switch ignition ON (engine off)	Neutral	Below 1.0	• Neutral switch No.2 • Related wiring harness
				Except above	B+	
2H	Ignition (IG1)	IG1 relay	Switch ignition ON (engine off)		Approx. 12.72	• IG1 relay • Related wiring harness
2I	—	—	—		—	—
2J*2	Back-up light	Back-up light switch	Shift lever is at R position		Below 1.0	• Back-up light switch • Related wiring harness
			Shift lever is not at R position		B+	
2K	Main relay control	Main relay	Switch ignition ON (engine off)		Approx. 0.85	• Main relay • Related wiring harness
2L	—	—	—		—	—
2M*2	CPP	CPP switch, start stop unit	Clutch pedal depressed		Below 1.0	• CPP switch • Start stop unit • Related wiring harness
			Clutch pedal released		B+	
2N	Selector lever position*1	TCM, start stop unit	Selector lever position is not P or N position		R position: Approx. 12.32	• TCM • Start stop unit • Related wiring harness
					M position: Approx. 12.45	
					D position: Approx. 12.48	
			Selector lever position is P or N position		P position: Approx. 0.17	
					N position: Approx. 0.17	
	Starter interlock*2	Starter interlock switch, start stop unit	Clutch pedal depressed		Below 1.0	• Starter interlock switch • Start stop unit • Related wiring harness
			Clutch pedal released		B+	
2O	Battery voltage	Battery	Switch ignition ON (engine off)		B+	• Related wiring harness

Terminal	Signal	Connected to	Test condition		Voltage (V)	Inspection item
2P	DC-DC converter control*4	DC-DC converter	Switch ignition ON (engine off)		Below 1.0	• DC-DC converter • Related wiring harness
	DC-DC converter (i-ELOOP) control*3	DC-DC converter (i-ELOOP)	Switch ignition ON (engine off)		Below 1.0	• DC-DC converter (i-ELOOP) • Related wiring harness
2Q	—	—	—		—	—
2R	—	—	—		—	—
2S	Battery voltage	Main relay	Switch ignition ON (engine off)		B+	• Related wiring harness
2T	Battery voltage	Main relay	Switch ignition ON (engine off)		B+	• Related wiring harness
2U	—	—	—		—	—
2V	GND	MAF sensor, IAT sensor No.1	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2W	Blow-by heater relay	Blow-by heater relay	Switch ignition ON (engine off)		Approx. 12.58	• Blow-by heater relay • Related wiring harness
			Idle (after warm up)		14.45—14.52	
2X	GND	GND	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2Y	—	—	—		—	—
2Z	GND	Refrigerant pressure sensor	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2AA	Brake (No.1)	Brake switch (No.1 signal)	Brake pedal released		Approx. 0.01	• Brake switch (No.1 signal) • Related wiring harness
			Brake pedal depressed		Approx. 12.47	
2AB	Brake (No.2)	Brake switch (No.2 signal)	Brake pedal released		Approx. 0.01	• Brake switch (No.2 signal) • Related wiring harness
			Brake pedal depressed		Approx. 12.12	
2AC	MAF	MAF sensor	Switch ignition ON (engine off)		Approx. 0.95	• MAF sensor • Related wiring harness
			Idle (after warm up)		1.34—1.36	
2AD	GND	GND	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2AE	GND	GND	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2AF	GND	GND	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2AG	Ambient temperature	Ambient temperature sensor	Switch ignition ON (engine off)		Approx. 2.06	• Ambient temperature sensor • Related wiring harness
2AH*2	GND	Clutch stroke sensor	Under any condition		Below 1.0	• Related wiring harness
2AI	Constant voltage (Vref)	MAF sensor	Switch ignition ON (engine off)		Approx. 5.01	• Related wiring harness
2AJ*2	Clutch stroke sensor	Clutch stroke sensor	Switch ignition ON (engine off)	Clutch pedal released	Approx. 4.50	• Clutch stroke sensor • Related wiring harness
				Clutch pedal depressed	Approx. 0.83	
2AK	Constant voltage (Vref)	Power brake unit vacuum sensor	Switch ignition ON (engine off)		Approx. 5.01	• Related wiring harness
2AL*2	Constant voltage (Vref)	Clutch stroke sensor	Switch ignition ON (engine off)		Approx. 5.01	• Related wiring harness
2AM	Constant voltage (Vref)	APP sensor No.1	Switch ignition ON (engine off)		Approx. 5.01	• Related wiring harness
2AN	APP (No.1)	APP sensor No.1	Switch ignition ON (engine off)	Accelerator pedal released	Approx. 0.80	• APP sensor No.1 • Related wiring harness
				Accelerator pedal depressed	Approx. 4.57	

Terminal	Signal	Connected to	Test condition		Voltage (V)	Inspection item
2AO	GND	APP sensor No.1	Under any condition		Approx. 0.01	• Related wiring harness
2AP	—	—	—		—	—
2AQ	A/C cut-off control	A/C relay	A/C relay OFF		Approx. 14.52	• A/C relay • Related wiring harness
			A/C relay ON		Approx. 0.20	
2AR	Constant voltage (Vref)	Refrigerant pressure sensor	Switch ignition ON (engine off)		Approx. 5.01	• Related wiring harness
2AS	IAT (No.1)	IAT sensor No.1	Switch ignition ON (engine off)		Approx. 1.96	• IAT sensor No.1 • Related wiring harness
2AT	GND	Ambient temperature sensor	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2AU*3	DC-DC converter (i-ELOOP) control	DC-DC converter (i-ELOOP)	Idle (after warm up)		0 or B+	• DC-DC converter (i-ELOOP) • Related wiring harness
2AV	—	—	—		—	—
2AW	Constant voltage (Vref)	APP sensor No.2	Switch ignition ON (engine off)		Approx. 5.01	• Related wiring harness
2AX	APP (No.2)	APP sensor No.2	Switch ignition ON (engine off)	Accelerator pedal released	Approx. 0.41	• APP sensor No.2 • Related wiring harness
				Accelerator pedal depressed	Approx. 2.29	
2AY	GND	APP sensor No.2	Switch ignition ON (engine off)		Approx. 0.01	• Related wiring harness
2AZ	Cooling fan control	Cooling fan relay	Cooling fan operating		Below 1.0	• Cooling fan relay • Related wiring harness
			Cooling fan not operating		B+	
2BA	—	—	—		—	—
2BB	—	—	—		—	—
2BC	Power brake unit vacuum	Power brake unit vacuum sensor	Switch ignition ON (engine off)	Brake pedal depressed (10 times)	Approx. 3.98	• Power brake unit vacuum sensor • Related wiring harness
			Idle (after warm up)	Brake pedal released	0.41—0.42	
2BD	GND	Power brake unit vacuum sensor	Switch ignition ON (engine off)		Approx. 0.32	• Related wiring harness
2BE	—	—	—		—	—

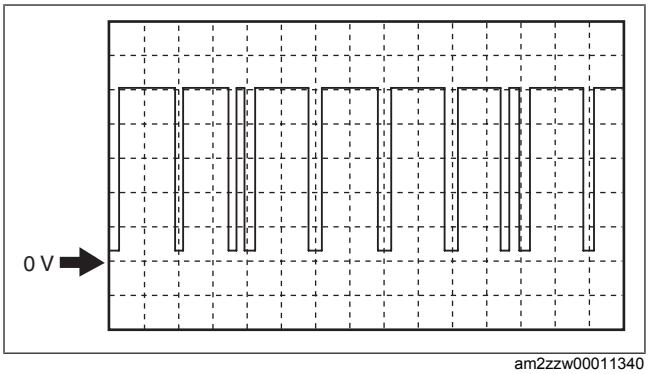
Terminal	Signal	Connected to	Test condition		Voltage (V)	Inspection item
2BF	Starter cut-off control	Starter relay, start stop unit	Switch ignition ON (engine off)	**MTX** • Clutch pedal released	B+	• Starter relay • Start stop unit • Related wiring harness
				ATX • Selector lever position is not P or N position	R position: Approx. 11.14	
					D position: Approx. 11.23	
					M position: Approx. 11.23	
				MTX • Clutch pedal depressed	Below 1.0	
				ATX • Selector lever position is P or N position	P position: Approx. 0.01	
					N position: Approx. 0.01	
2BG	Fan control	Fan control module	Switch ignition ON (engine off)		Approx. 0.61	• Fan control module • Related wiring harness
2BH	Refrigerant pressure	Refrigerant pressure sensor	Switch ignition ON (engine off)	A/C switch off	Approx. 0.94	• Refrigerant pressure sensor • Related wiring harness
			Idle (after warm up) or switch ignition ON (engine off)		1.39—1.59	

*1 : ATX
*2 : MTX
*3 : With i-ELOOP
*4 : Without i-ELOOP
*5 : With engine oil level sensor

Inspection Using An Oscilloscope (Reference)
CMP (G+) signal
PCM terminals
- 1A(+)—body ground(–)

Oscilloscope setting
- 1 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

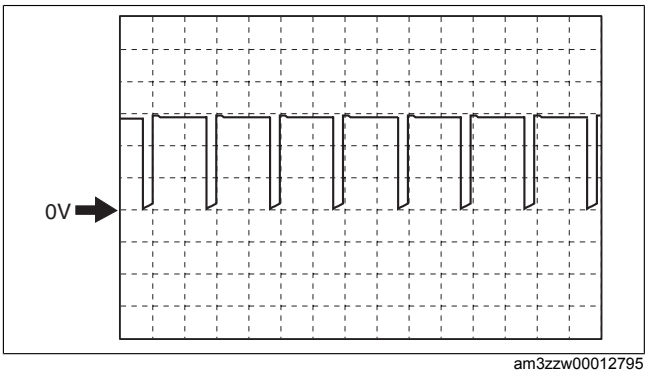


am2zzw00011340

Intake shutter valve (ISV-) signal
PCM terminals
- 1AS(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

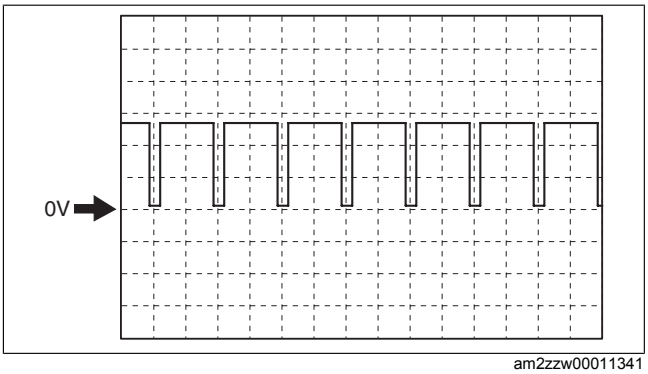


am3zzw00012795

LP-EGR control valve (LPEGR-)
PCM terminals
- 1AI(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 µs/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

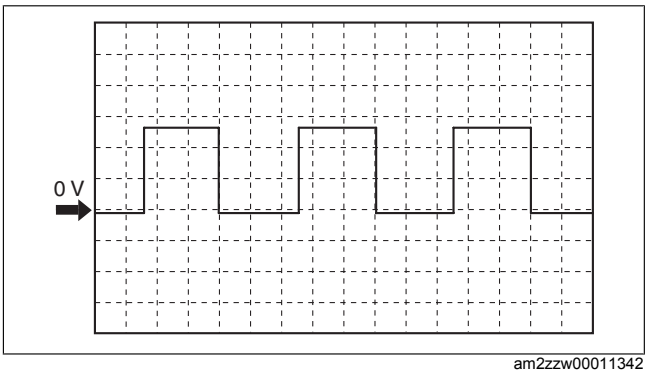


am2zzw00011341

Suction control valve signal
PCM terminals
- 1CC(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

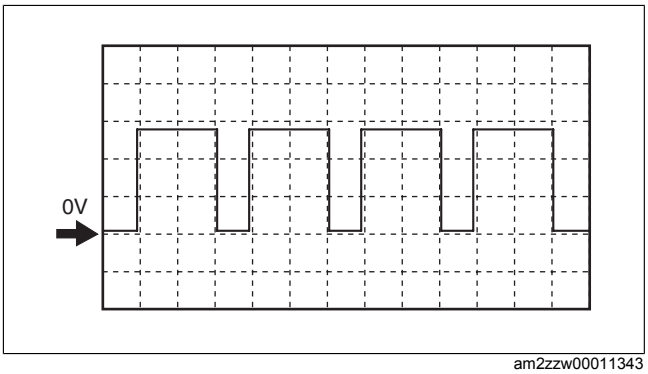


am2zzw00011342

A/F sensor heater control signal
PCM terminals
- 1AD(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

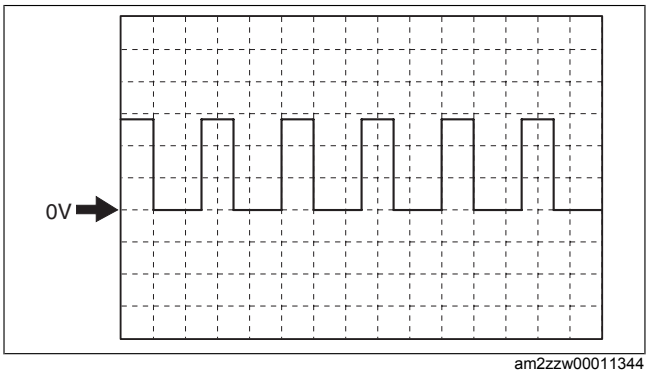


am2zzw00011343

Engine oil control signal
PCM terminals
- 1CO(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

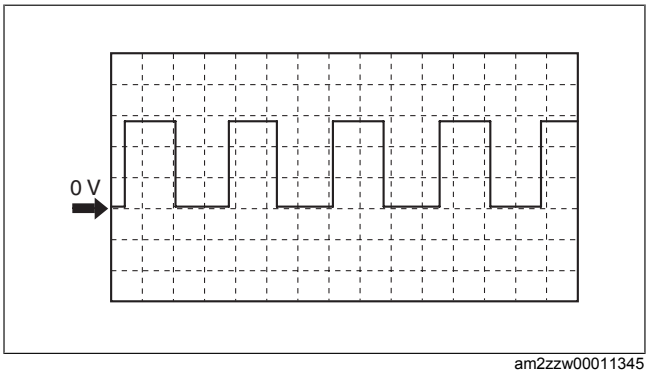


am2zzw00011344

Turbocharger solenoid valve
PCM terminals
- 1CH(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

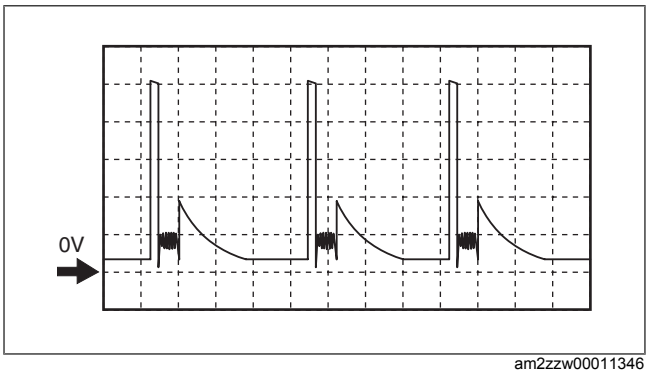


am2zzw00011345

Fuel injection control (-) signal
PCM terminals
- Fuel injector No.1: 1EA(+)—body ground(–)
- Fuel injector No.2: 1CW(+)—body ground(–)
- Fuel injector No.3: 1DA(+)—body ground(–)
- Fuel injector No.4: 1DW(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 400 µs/DIV (X), DC range

Vehicle condition
- Idle after warm up

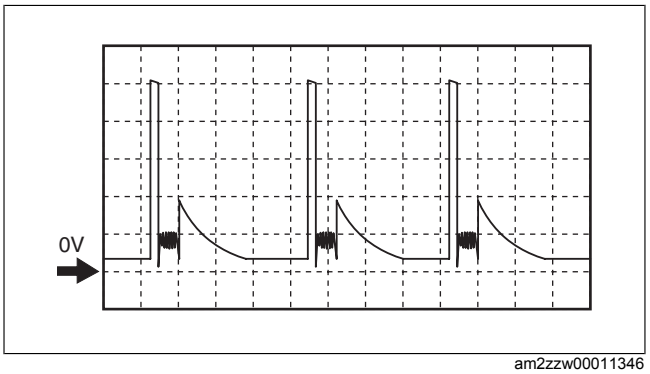


am2zzw00011346

Fuel injection control (+) signal
PCM terminals
- Fuel injector No.1: 1EE(+)—body ground(–)
- Fuel injector No.2: 1CT(+)—body ground(–)
- Fuel injector No.3: 1CS(+)—body ground(–)
- Fuel injector No.4: 1EF(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 400 µs/DIV (X), DC range

Vehicle condition
- Idle after warm up

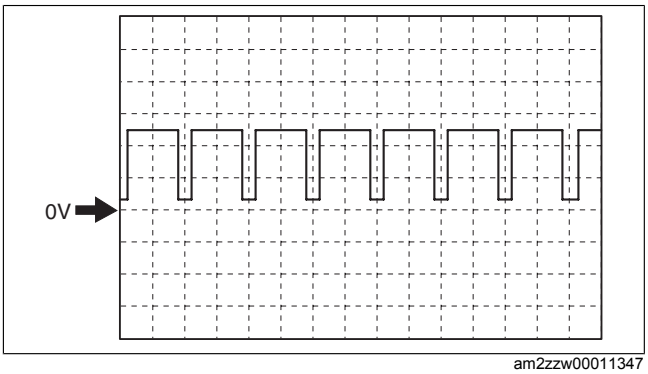


am2zzw00011346

Generator output voltage signal
PCM terminals
- 1BU(+)—body ground(–) (without i-ELOOP)
- 1O(+)—body ground(–) (with i-ELOOP)

Oscilloscope setting
- 2 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

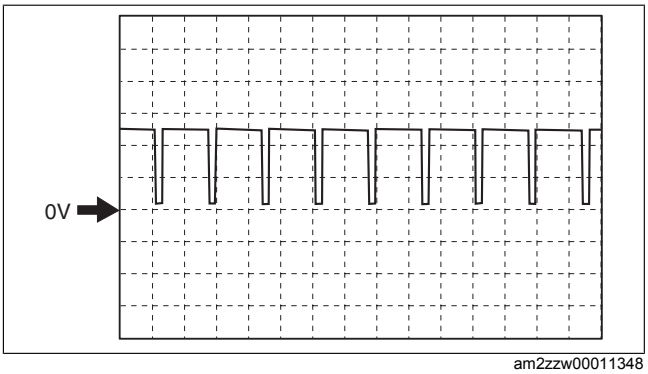


am2zzw00011347

CKP (NE+) signal
PCM terminals
- 1E(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

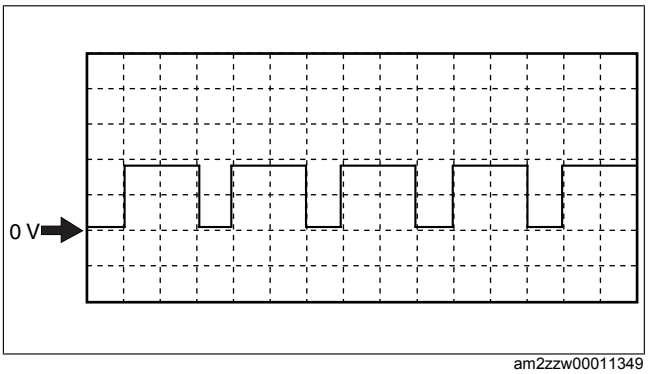


am2zzw00011348

Generator field coil control signal
PCM terminals
- 1AY(+)—body ground(–) (without i-ELOOP)
- 1CG(+)—body ground(–) (with i-ELOOP)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

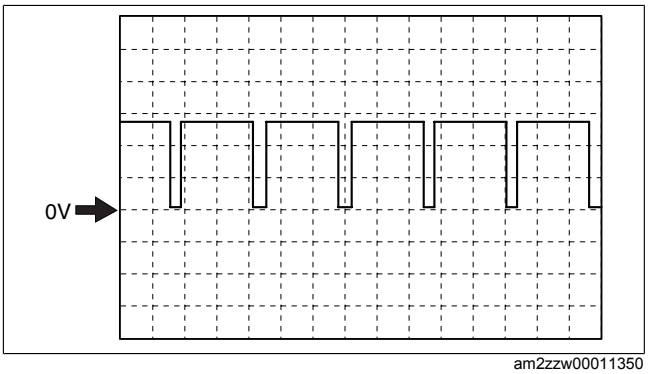


am2zzw00011349

Exhaust shutter valve (ESV-)
PCM terminals
- 1AJ(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 400 µs/DIV (X), DC range

Vehicle condition
- Idle after warm up



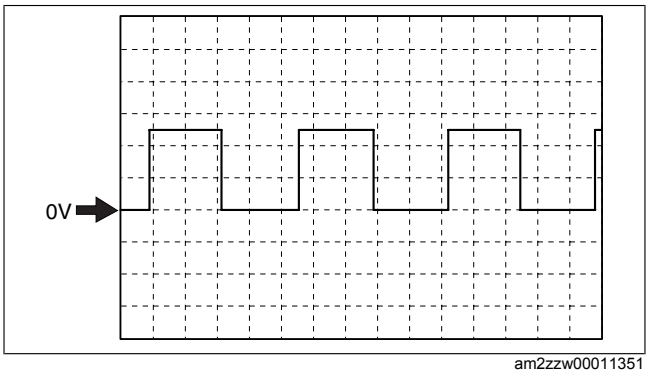
am2zzw00011350

TURBINE ROTATION SPEED SIGNAL

PCM terminals
- 1EJ(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 10 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



am2zzw00011351

Using The M-MDS

Note
- PIDs for the following parts are not available on this model. Go to the appropriate part inspection page.
  - — CMP sensor (See CAMSHAFT POSITION (CMP) SENSOR INSPECTION [SKYACTIV-D 1.5].)
  - — Main relay (See RELAY INSPECTION.)

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Measure the PID value.
   - If PID value is not within the specification, follow the instructions in action column.

Note
- The PID/DATA MONITOR function monitors the calculated value of the input/output signals in the PCM. Therefore, an output device malfunction is not directly indicated as a malfunction of the monitored value for the output device. If a monitored value of an output device is out of specification, inspect the monitored value of the input device related to the output control.
- The simulation items that are used in the engine control system operation inspection are as follows.
  - — ACCS, ARPMDES, CCV_DSD, E_WP, EXHSHUT_DSD, FAN_DUTY, HPEGR_DSD, INJ_1, INJ_2, INJ_3, INJ_4, ISV_DSD, LPEGR_DSD, VGTURB_DSD

PID/DATA monitor item table

—: Not applicable

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
AC_PRES	kPa, Bar, psi	Refrigerant pressure	Converted value	**Idle** • A/C switch ON: Approx. 930 kPa {9.48 kgf/cm^2, 135 psi}	2BH
AC_REQ	Off/On	A/C request signal	Converted value	• A/C switch OFF: Off • A/C switch ON: On	CAN(2C, 2D)
ACCS	Off/On	A/C relay	Output	• A/C relay is OFF: Off • A/C relay is ON: On	2AQ
ALTT V	V	Generator output voltage	Converted value*3, Input*4	• Switch ignition ON (engine off): 0 V • Idle: Approx. 13 V • Racing (engine speed 2,000 rpm): Approx. 9.6 V	1O*3, 1BU*4
AMB_TEMP	°C, °F	Ambient air temperature	Converted value	• Displays the ambient air temperature	2AG
APP1	V	APP sensor No.1 voltage	Input	**Switch ignition ON (engine off)** • Accelerator pedal released: Approx. 0.79 V • Accelerator pedal depressed: Approx. 4.54 V	2AN
	%	APP sensor No.1	Converted value	**Switch ignition ON (engine off)** • Accelerator pedal released: Approx. 0% • Accelerator pedal depressed: Approx. 100%	

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
APP2	V	APP sensor No.2 voltage	Input	**Switch ignition ON (engine off)** • Accelerator pedal released: Approx. 0.39 V • Accelerator pedal depressed: Approx. 2.27 V	2AX
	%	APP sensor No.2	Converted value	**Switch ignition ON (engine off)** • Accelerator pedal released: Approx. 0% • Accelerator pedal depressed: Approx. 100%	
ARPMDES	RPM	Target engine speed	Calculation	• Displays the target engine speed	—
BARO	kPa, Bar, psi	Barometric pressure	Converted value	• Displays BARO	—
BATT_CUR	A	Battery current	Input	• Displays the battery charge/discharge current value	CAN(2C, 2D)
BATT_DAY	—	Number of days elapsed since current sensor initialization	Calculation	• Displays the vehicle battery days in service	CAN(2C, 2D)
BATT_RES	—	Battery internal resistance (estimated)	Calculation	• Displays the battery inferred internal resistance	CAN(2C, 2D)
BATT_SOC	%	Battery charge condition (estimated)	Calculation	• Displays the battery estimated state of charge	CAN(2C, 2D)
BATT_TEMP	°C, °F	Battery temperature	Input	• Displays the battery fluid temperature	CAN(2C, 2D)
BATT_V	V	Battery voltage	Input	• Displays the battery voltage	CAN(2C, 2D)
BBP	kPa, Bar, psi	Power brake unit vacuum sensor	Calculation	• Switch ignition ON (engine off) and depress the brake pedal 10 times: Approx. 97.85 kPa {0.9978 kgf/cm^2, 14.19 psi} • Idle: Approx. 7.97 kPa {0.0831 kgf/cm^2, 1.16 psi} • The instant the brake pedal is depressed and released while idling: Approx. 25.46 kPa {0.2596 kgf/cm^2, 3.693 psi}	2BC
BFP	kPa, Bar, psi	Brake fluid pressure	Input	**Idle** • Brake pedal released: 0 kPa {0 kgf/cm^2, 0 psi} • Brake pedal depressed: Approx. 10.43 kPa {0.1064 kgf/cm^2, 1.513 psi}	CAN
BOO	High/Low	Brake switch (No.1 signal)	Converted value	• Brake pedal released: Low • Brake pedal depressed: High	2AA
BPA	High/Low	Brake switch (No.2 signal)	Converted value	• Brake pedal released: Low • Brake pedal depressed: High	2AB
CACT11	°C, °F	Boost air temperature	Converted value	• Displays the boost air temperature	2N
CC_DIFP_WOA	°(deg)	Actual difference in phase between camshaft and crankshaft (no correction)	Calculation	• Displays the actual difference in phase between camshaft and crankshaft	—
CCV_DSD	°(deg)	Target coolant control valve opening angle	Calculation	• Displays the target coolant control valve opening angle	—
CCV_ACT	°(deg)	Actual coolant control valve opening angle	Converted value	• Displays the actual coolant control valve opening angle	1BP
CLR_DIST	—	Distance after DTC cleared	Calculation	• Displays the mileage after DTC cleared	—
CLU_CUT_SW*1	Off/On	Starter interlock switch	Converted value	• Starter interlock switch ON: On • Starter interlock switch OFF: Off	2N
CLU_SW*1	Off/On	CPP switch	Converted value	• Clutch pedal depressed: On • Clutch pedal released: Off	2M
CPP*1	%	Clutch pedal position	Converted value	• Displays the clutch pedal position	2AJ

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
CPP/PNP*1	Off/On	Shift lever position	Converted value	• Neutral: On • Other than neutral: Off	2B, 2G
DPF_LMP	Off/On	Diesel particulate filter indicator light	Output	• Diesel particulate filter indicator light not illuminate: Off • Diesel particulate filter indicator light illuminate: On	—
DPF_LMP_CNT	—	Number of times diesel particulate filter indicator light illuminates	Calculation	• Displays the number of times diesel particulate filter indicator light illuminates	—
DPF_REG_CNT	—	Diesel particulate filter regeneration count	Calculation	• Displays the diesel particulate filter regeneration count	—
E_WP	%	Electric water pump control duty value	Output	• Displays the electric water pump control duty value	1CP
ECT	°C, °F	Engine coolant temperature	Converted value	• Displays ECT	1CY
EOL*5	mm, in	Engine oil level	Converted value	• Displays the engine oil level	1Q
EOP	kPa, Bar, psi	Engine oil pressure	Converted value	• Switch ignition ON (engine off): Approx. 1 kPa {0.01 kgf/cm^2, 0.1 psi} • Idle: Approx. 141 kPa {1.44 kgf/cm^2, 20.5 psi} • Racing (engine speed 4,000 rpm): Approx. 390 kPa {3.98 kgf/cm^2, 56.6 psi}	1AB
EOT	°C, °F	Engine oil temperature	Converted value	• Displays the engine oil temperature	1W
EOT2*5	°C, °F	Engine oil temperature (engine oil level sensor)	Converted value	• Displays the engine oil temperature	1Q
EXHPRES1	kPa, Bar, psi	Exhaust gas pressure (No. 1)	Converted value	• Idle: Approx. 100 kPa {1.02 kgf/cm^2, 14.5 psi} • Racing (engine speed above 2,000 rpm): Approx. 115 kPa {1.17 kgf/cm^2, 16.7 psi} • Racing (engine speed above 4,000 rpm): Approx. 205 kPa {2.09 kgf/cm^2, 29.7 psi}	1X
EXHPRESS_DIF	kPa, Bar, psi	Exhaust gas pressure (No. 2)	Converted value	• Displays the difference in pressure between exhaust gas pressure before and after passing the diesel particulate filter	1AC
EXHSHUT_ACT	%	Exhaust shutter valve control actual value	Converted value	• Displays the exhaust shutter valve control actual value	1AL
EXHSHUT_DSD	%	Exhaust shutter valve control desired value	Calculation	• Displays the exhaust shutter valve control desired value	—
EXHTEMP1	°C, °F	Exhaust gas temperature (No.1)	Converted value	• Displays the exhaust gas temperature (No.1)	1CI
EXHTEMP2	°C, °F	Exhaust gas temperature (No.2)	Converted value	• Displays the exhaust gas temperature (No.2)	1CA
EXHTEMP3	°C, °F	Exhaust gas temperature (No.3)	Converted value	• Displays the exhaust gas temperature (No.3)	1BW
EXHTEMP4	°C, °F	Exhaust gas temperature (No.4)	Converted value	• Displays the exhaust gas temperature (No.4)	1CE
FAN_DUTY	%	Fan control module	Calculation	**Idle** • ECT is below 100 °C {212 °F}: 0 % • ECT is above 100 °C {212 °F}: Approx. 30—100 % (after a certain period has elapsed from when ECT reaches 100 °C {212 °F})	2BG

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
FI_LRN_11	—(µs)	Fuel injection learning value (fuel injector No.1 at 35 MPa {357 kgf/cm^2, 5,076 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.1 at 35 MPa {357 kgf/cm^2, 5,076 psi})	—
FI_LRN_12	—(µs)	Fuel injection learning value (fuel injector No.2 at 35 MPa {357 kgf/cm^2, 5,076 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.2 at 35 MPa {357 kgf/cm^2, 5,076 psi})	—
FI_LRN_13	—(µs)	Fuel injection learning value (fuel injector No.3 at 35 MPa {357 kgf/cm^2, 5,076 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.3 at 35 MPa {357 kgf/cm^2, 5,076 psi})	—
FI_LRN_14	—(µs)	Fuel injection learning value (fuel injector No.4 at 35 MPa {357 kgf/cm^2, 5,076 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.4 at 35 MPa {357 kgf/cm^2, 5,076 psi})	—
FI_LRN_21	—(µs)	Fuel injection learning value (fuel injector No.1 at 65 MPa {663 kgf/cm^2, 9,427 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.1 at 65 MPa {663 kgf/cm^2, 9,427 psi})	—
FI_LRN_22	—(µs)	Fuel injection learning value (fuel injector No.2 at 65 MPa {663 kgf/cm^2, 9,427 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.2 at 65 MPa {663 kgf/cm^2, 9,427 psi})	—
FI_LRN_23	—(µs)	Fuel injection learning value (fuel injector No.3 at 65 MPa {663 kgf/cm^2, 9,427 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.3 at 65 MPa {663 kgf/cm^2, 9,427 psi})	—
FI_LRN_24	—(µs)	Fuel injection learning value (fuel injector No.4 at 65 MPa {663 kgf/cm^2, 9,427 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.4 at 65 MPa {663 kgf/cm^2, 9,427 psi})	—
FI_LRN_31	—(µs)	Fuel injection learning value (fuel injector No.1 at 95 MPa {969 kgf/cm^2, 13,779 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.1 at 95 MPa {969 kgf/cm^2, 13,779 psi})	—
FI_LRN_32	—(µs)	Fuel injection learning value (fuel injector No.2 at 95 MPa {969 kgf/cm^2, 13,779 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.2 at 95 MPa {969 kgf/cm^2, 13,779 psi})	—
FI_LRN_33	—(µs)	Fuel injection learning value (fuel injector No.3 at 95 MPa {969 kgf/cm^2, 13,779 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.3 at 95 MPa {969 kgf/cm^2, 13,779 psi})	—
FI_LRN_34	—(µs)	Fuel injection learning value (fuel injector No.4 at 95 MPa {969 kgf/cm^2, 13,779 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.4 at 95 MPa {969 kgf/cm^2, 13,779 psi})	—
FI_LRN_41	—(µs)	Fuel injection learning value (fuel injector No.1 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.1 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	—
FI_LRN_42	—(µs)	Fuel injection learning value (fuel injector No.2 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.2 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	—

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
FI_LRN_43	—(μs)	Fuel injection learning value (fuel injector No.3 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.3 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	—
FI_LRN_44	—(μs)	Fuel injection learning value (fuel injector No.4 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.4 at 125 MPa {1,275 kgf/cm^2, 18,130 psi})	—
FI_LRN_51	—(μs)	Fuel injection learning value (fuel injector No.1 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.1 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	—
FI_LRN_52	—(μs)	Fuel injection learning value (fuel injector No.2 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.2 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	—
FI_LRN_53	—(μs)	Fuel injection learning value (fuel injector No.3 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.3 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	—
FI_LRN_54	—(μs)	Fuel injection learning value (fuel injector No.4 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	Calculation	• Displays the fuel injection learning value (fuel injector No.4 at 160 MPa {1,632 kgf/cm^2, 23,206 psi})	—
FIA_DSD	—(mm^3/ Stroke)	Supply pump flow desired value	Calculation	• Displays the supply pump flow desired value	—
FIP_FL	A	Supply pump flow control current	Output	• Switch ignition ON (engine off): 0 A • Idle: Approx. 1.8 A • Racing (engine speed above 4,000 rpm): Approx. 1.7 A	—
FIP_SCV	A	Suction control valve	Output	• Switch ignition ON (engine off): Approx. 1.03 A • Idle: Approx. 1.85 A • Racing (engine speed above 4,000 rpm): Approx. 1.74 A	1BY, 1CC
FP_DUTY	%	Supply pump duty cycle	Calculation	• Switch ignition ON (engine off): 0 % • Idle: Approx. 45 % • Racing (engine speed above 4,000 rpm): Approx. 42 %	—
FP_RCV	—(ms)	Fuel pressure relief valve operation time	Calculation	• Switch ignition ON (engine off): 0 • Idle: 0 • Racing (engine speed above 4,000 rpm): 0	1DG, 1DK
FRP	kPa, Bar, psi	Common rail pressure	Converted value	• Switch ignition ON (engine off): 360 kPa {3.67 kgf/cm^2, 52.2 psi} • Idle: Approx. 40 MPa {408 kgf/cm^2, 5802 psi} • Racing (engine speed above 4,000 rpm): Approx. 87.3 MPa {890 kgf/cm^2, 12662 psi}	1AG
FRP_DSD	kPa, Bar, psi	Common rail pressure desired value	Calculation	• Displays the common rail pressure desired value	—
FRT	°C, °F	Fuel temperature inside the fuel supply line	Converted value	• Displays the fuel temperature inside the fuel supply line	—
GENVDSD	V	Generator field coil control desired value	Calculation	• Switch ignition ON (engine off): Approx. 12.8 V • Idle: Approx. 14.5 V	—
GLOW_P_V	V	Glow plug voltage control desired value	Calculation	• Displays the glow plug voltage control desired value	—

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
GP_LMP	Off/On	Glow indicator light	Calculation	• Switch ignition ON (engine off): Off • Idle: Off	2B, 2G
HPEGR_ACT	%	HP-EGR control valve position control actual value	Converted value	• Switch ignition ON (engine off): 0 % • Idle: Approx. 28.6 % • Racing (engine speed 2,000 rpm): Approx. 39.6 % • Racing (engine speed 4,000 rpm): Approx. 0 %	1BF
HPEGR_DSD	%	HP-EGR control valve position control desired value	Calculation	• Switch ignition ON (engine off): 0 % • Idle: Approx. 28.6 % • Racing (engine speed 2,000 rpm): Approx. 39.6 % • Racing (engine speed 4,000 rpm): Approx. 0 %	—
HTR11	%	A/F sensor heater control	Output	• Switch ignition ON (engine off): 0 % • Idle: Approx. 22.74 % • Racing (engine speed above 4,000 rpm): Approx. 38.43 %	1AD
IAT	°C, °F	Intake air temperature (No. 1)	Converted value	• Displays the intake air temperature (No. 1)	2AS
INGEAR*2	Off/On	Gears are engaged	Calculation	• Selector lever at R, D or M position: On • Selector lever at P or N position: Off	CAN(2C, 2D)
INJ_AL_DIS	km, ft, mi	Distance travelled when automatic fuel injection amount learning	Calculation	• Displays the distance travelled when automatic fuel injection amount learning	—
INJ_AL_FRQ	—	Number of times automatic fuel injection amount learning is completed	Calculation	• Displays the number of times automatic fuel injection amount learning is completed	—
INJ_WL_DIS	km, ft, mi	Distance travelled when fuel injection amount learning at service factory	Calculation	• Displays the distance travelled when fuel injection amount learning at service factory	—
INJ_WL_FRQ	—	Number of times fuel injection amount learning is completed at service factory	Calculation	• Displays the number of times fuel injection amount learning is completed at service factory	—
INJ1_CMP	—(mm^3/ Stroke)	Fuel injector No.1 correction value	Calculation	• Displays the fuel injector No.1 correction value	—
INJ2_CMP	—(mm^3/ Stroke)	Fuel injector No.2 correction value	Calculation	• Displays the fuel injector No.2 correction value	—
INJ3_CMP	—(mm^3/ Stroke)	Fuel injector No.3 correction value	Calculation	• Displays the fuel injector No.3 correction value	—
INJ4_CMP	—(mm^3/ Stroke)	Fuel injector No.4 correction value	Calculation	• Displays the fuel injector No.4 correction value	—
I-Stop_OFF	Off/On	i-stop OFF mode	Converted value	• i-stop OFF switch OFF: Off • i-stop OFF switch ON: On	—
I-Stop_TRD	Off/On	i-stop transmission D position selected status	Calculation	• D position:On • Except above: Off	—
I-Stop_VSP	Off/On	i-stop vehicle speed history flag	Calculation	• Vehicle speed in which engine stop condition is met via i-stop control is detected: On • Except above: Off	—
I-Stop_VST	Off/On	i-stop vehicle stop flag	Calculation	• Vehicle stop predicted: On • Except above: Off	—
ISV_ACT	°(deg)	Intake shutter valve control actual value	Calculation	• Switch ignition ON (engine off): Approx. 88.28 • Idle: Approx. 4.36 • Racing (engine speed above 4,000 rpm): Approx. 82.5	1AX
ISV_DSD	°(deg) %	Intake shutter valve control desired value	Calculation	• Displays the intake shutter valve control desired value	—

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
ISV_LRN_C	°(deg)	Intake shutter valve learning value (closed)	Calculation	• Displays the intake shutter valve learning value (closed)	—
ISV_POS	%	Intake shutter valve	Converted value	• Switch ignition ON (engine off): Approx. 92 % • Idle: Approx. 3 % • Racing (engine speed 2,000 rpm): Approx. 18 % • Racing (engine speed 4,000 rpm): Approx. 81.9 %	1AV
LOAD	%	Engine load	Calculation	• Idle: Approx. 16 % • Racing (engine speed above 2,000 rpm): Approx. 4.3 % • Racing (engine speed above 5,000 rpm): Approx. 8.6 %	—
LP_EGR_ACT	%	LP-EGR control valve position control actual value	Converted value	• Switch ignition ON (engine off): 0 % • Idle: 0 % • Racing (engine speed 2,000 rpm): 0 % • Racing (engine speed 4,000 rpm): Approx. 0 %	1AQ
LPEGR_DSD	%	LP-EGR control valve position control desired value	Calculation	• Displays the LP-EGR control valve position control desired value	—
M_GEAR*1	Neutral/ 1st gear/ 2nd gear/ 3rd gear/ 4th gear/ 5th gear/ 6th gear/ Reverse/ Undefined /Auto/ In_Progress/YSF/ Error	Manual gear position	Calculation	• Displays the manual gear position	CAN(2C, 2D)
MAF	g/Sec	Mass air flow	Converted value	• Switch ignition ON (engine off): Approx. 0.40 g/s {0.053 lb/min} • Idle: Approx. 3.20 g/s {0.423 lb/min} • Racing (engine speed 2,000 rpm): Approx. 9.70 g/s {1.28 lb/min} • Racing (engine speed 4,000 rpm): Approx. 62.8 g/s {8.31 lb/min}	2AC
MAP_DSD	kPa, Bar, psi	Manifold absolute pressure (No.2) desired value	Calculation	• Displays the manifold absolute pressure (No.2) desired value	—
MIL	Off/On	Check engine light	Calculation	• Check engine light not illuminate: Off • Check engine light illuminate: ON	—
MIL_DIS	km, ft, mi	Travelled distance since check engine light illuminated	Calculation	• Displays the travelled distance since check engine light illuminated	—
O2	%	Oxygen concentration in exhaust gas	Calculation	• Switch ignition ON (engine off): 0 % • Idle: 14 % • Racing (engine speed above 4,000 rpm): Approx. 16 %	1H, 1I
O2S_IMP	Ω	A/F sensor element impedance	Converted value	• Displays the A/F sensor element impedance	—

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
O2S11	μA	A/F sensor current	Input	• Idle: Approx. 990 μA • Racing (engine speed 2,000 rpm): Approx. 1.12 mA	1H, 1I
	V	A/F sensor voltage		• Switch ignition ON (engine off): 2.22 V • Idle: Approx. 2.8 V • Racing (engine speed 2,000 rpm): Approx. 2.9 V • Racing (engine speed 4,000 rpm): Approx. 3.0 V	
O2S11_CAL	—	A/F sensor calibration value	Calculation	• Switch ignition ON (engine off): Approx. 0.97 • Idle: Approx. 0.97 • Racing (engine speed 4,000 rpm): Approx. 0.97	—
O2S11_MODE	—	A/F sensor activation status	Calculation	• Switch ignition ON (engine off): 0 • Idle: 3 • Racing (engine speed 2,000 rpm): 3	—
OIL_DIL	kg, lb	Engine oil dilution amount	Calculation	• Displays the engine oil dilution amount	—
OIL_P_DUTY	%	Engine oil pressure control circuit duty cycle	Output	• Switch ignition ON (engine off): 0 % • Idle: Approx. 54 % • Racing (engine speed 4,000 rpm): 0 %	1CO
OILCHG_DIS	km, ft, mi	Distance from the last engine oil change	Calculation	• Displays the distance from the last engine oil change	—
PCVHC	%	Blow-by heater control	Output	**ECT: above 90 °C {194 °F}** • Switch ignition ON (engine off): 0 % • Idle: 0 %	2W
PM_ACC	—(g/l, lb/ ft^3)	PM accumulation amount	Calculation	• Displays the PM accumulation amount	—
PM_ACC_DSD	—(g/l, lb/ ft^3)	PM accumulation amount desired	Calculation	• Displays the PM accumulation amount desired	—
PM_GEN	—(g/l, lb/ ft^3)	PM generation amount	Calculation	• Displays the PM generation amount	—
PN_SW*1	Open/ Closed	Park/Neutral position determination	Converted value	• Selector lever at R, D or M position: Open • Selector lever at P or N position: Open	CAN(2C, 2D)
REG_DIS	km, ft, mi	Distance since last diesel particulate filter regeneration	Calculation	• Displays the distance since last diesel particulate filter regeneration	—
REV_SW*1	Off/On	Reverse position determination	Calculation	• Displays the reverse position	CAN(2C, 2D)
RPM	RPM	Engine speed	Converted value	• Displays the engine speed	1E, 1F
VGTURB_ACT	%	Turbocharger actuator position control actual value	Converted value	• Displays the turbocharger actuator position control actual value	1BK
VGTURB_DSD	%	Turbocharger actuator position control desired value	Input	• Displays the turbocharger actuator position control desired value	—
VGTURB_RPM	RPM	Variable geometry turbocharger turbine rotation speed	Converted value	• Displays the variable geometry turbocharger turbine rotation speed	1EJ
VPWR	V	Battery positive voltage	Input	• Switch ignition ON (engine off): Approx. 12.78 V • Idle: Approx. 13.78 V	2O
VSS	KPH, MPH	Vehicle speed	Calculation	• Displays the vehicle speed	—

*1 : MTX
*2 : ATX
*3 : With i-ELOOP

*4 : Without i-ELOOP
*5 : With engine oil level sensor